Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Physics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Simple Harmonic Motion

Unit 2: Waves

Unit 3: Geometrical Optics

Unit 4: Electrostatics

Unit 5: Current Electricity

Unit 6: Electromagnetism

Unit 7: Basic Electronics

Unit 8: Atomic and Nuclear Physics

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Unit 1: Simple Harmonic Motion

Learning Objectives

• Define vibratory motion and SHM with examples.

• Explain the terms: amplitude, time period, frequency.

• Describe the simple pendulum and its time period.

• Solve numericals using T = 2 pi sqrt(l/g).

Key Concepts

• Vibratory motion: to-and-fro about mean position

• SHM: acceleration proportional to displacement, opposite in direction

• Amplitude, time period, frequency (f = 1/T)

• Simple pendulum and its laws

• Damped oscillations and resonance idea

Teaching Methodology (Step by Step)

1. Swing a pendulum and ask what repeats: introduce vibratory motion.
2. Define SHM with the mass-spring example on the board.
3. Demonstrate how length changes the pendulum's period.
4. Solve numericals on the pendulum formula together.

Classroom Activities

• Measure the period of pendulums of different lengths; plot results.

• Find examples of SHM at home (swing, fan spring).

Assessment Questions

1. Define simple harmonic motion.
2. What is the relation between frequency and time period?
3. A pendulum has length 1 m. Find its period (g = 10 m/s^2).
4. How does the period change if the length is quadrupled?

Homework

• Solve numericals on the simple pendulum.

• Write definitions of amplitude, period and frequency.

Time Allocation

5 periods of 40 minutes each

Unit 2: Waves

Learning Objectives

• Define a wave and distinguish mechanical and electromagnetic waves.

• Distinguish transverse and longitudinal waves.

• Define wavelength, frequency, speed and relate them (v = f lambda).

• Explain the uses of sound waves and ultrasound.

Key Concepts

• Wave: disturbance transferring energy

• Mechanical vs electromagnetic waves

• Transverse (light) vs longitudinal (sound)

• Wave equation: v = f lambda

• Reflection, refraction of waves; ultrasound uses

Teaching Methodology (Step by Step)

1. Drop a stone in water: watch ripples carry energy, not water.
2. Show transverse waves with a rope, longitudinal with a slinky.
3. Derive v = f lambda with a diagram.
4. Discuss ultrasound: medical scans, SONAR.

Classroom Activities

• Make waves in a rope and measure wavelength roughly.

• List five uses of ultrasound in pairs.

Assessment Questions

1. What is the difference between transverse and longitudinal waves?
2. Write the wave equation and define each term.
3. A wave has f = 50 Hz and lambda = 2 m. Find its speed.
4. Name two uses of ultrasound.

Homework

• Solve numericals on the wave equation.

• Draw transverse and longitudinal waves with labels.

Time Allocation

6 periods of 40 minutes each

Unit 3: Geometrical Optics

Learning Objectives

• State the laws of reflection.

• Describe image formation by plane and spherical mirrors.

• Explain refraction and total internal reflection.

• Describe image formation by convex and concave lenses.

Key Concepts

• Reflection: angle i = angle r

• Mirrors: plane, concave and convex; ray diagrams

• Refraction: Snell's law idea; refractive index

• Total internal reflection: fibre optics

• Lenses: convex (converging), concave (diverging)

Teaching Methodology (Step by Step)

1. Shine a torch on a mirror: trace incident and reflected rays.

2. Draw ray diagrams for mirrors step by step; students copy.

3. Show refraction with a pencil in a glass of water.

4. Demonstrate total internal reflection with a laser and water stream.

Classroom Activities

• Draw ray diagrams for all mirror cases in pairs.

• Find the focal length of a convex lens using sunlight.

Assessment Questions

1. State the laws of reflection.

2. Draw the ray diagram for an object beyond 2F of a concave mirror.

3. What is total internal reflection? Give one use.

4. Distinguish between a convex and a concave lens.

Homework

• Draw all ray diagrams for mirrors and lenses.

• Write two uses of fibre optics.

Time Allocation

7 periods of 40 minutes each

Unit 4: Electrostatics

Learning Objectives

• Explain charging by rubbing, conduction and induction.

• State Coulomb's law and solve simple numericals.

• Define electric field and electric potential.

• Explain capacitors and their uses.

Key Concepts

• Static charge: positive and negative

• Charging methods: friction, conduction, induction

• Coulomb's law: F = k q1 q2 / r^2

• Electric field lines and their properties

• Capacitor: construction, capacitance, uses

Teaching Methodology (Step by Step)

1. Rub a balloon on hair and pick up paper bits: the hook.
2. Demonstrate attraction and repulsion with charged rods.
3. Explain Coulomb's law parallel to gravitation.
4. Show field-line patterns for single and double charges.

Classroom Activities

• Charge combs and test attraction on paper pieces.

• Draw field lines for like and unlike charges.

Assessment Questions

1. Name three methods of charging a body.
2. State Coulomb's law.
3. Draw electric field lines around a positive charge.
4. What is a capacitor? Give one use.

Homework

• Solve numericals on Coulomb's law.

• Draw field-line patterns in the notebook.

Time Allocation

6 periods of 40 minutes each

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Unit 5: Current Electricity

Learning Objectives

• Define electric current and its unit.

• State Ohm's law and solve numericals (V = IR).

• Distinguish series and parallel circuits.

• Explain electric power and calculate energy cost.

Key Concepts

• Current: flow of charge; unit ampere

• Ohm's law: V = IR; resistance and its unit

• Factors affecting resistance

• Series vs parallel: current, voltage, equivalent resistance

• Power: P = VI; kilowatt-hour and billing

Teaching Methodology (Step by Step)

1. Ask why a bulb glows: build the current idea.

2. Demonstrate Ohm's law with a simple circuit and meters.

3. Compare series and parallel bulbs: brightness tells the story.

4. Calculate the cost of running a 100 W bulb for 10 hours.

Classroom Activities

• Build series and parallel circuits with cells and bulbs.

• Read a home electricity bill and identify the units consumed.

Assessment Questions

1. State Ohm's law.

2. A bulb draws 2 A at 220 V. Find its resistance.

3. Why are home appliances connected in parallel?

4. Calculate the energy used by a 60 W bulb in 5 hours (in kWh).

Homework

• Solve numericals on Ohm's law and power.

• Draw series and parallel circuits with labels.

Time Allocation

6 periods of 40 minutes each

Unit 6: Electromagnetism

Learning Objectives

• Describe the magnetic field around a current-carrying wire.

• Explain the force on a current-carrying conductor (Fleming's left-hand rule).

• Describe electromagnetic induction (Faraday's law).

• Explain the working of a transformer.

Key Concepts

• Oersted's discovery: current makes magnetism

• Right-hand grip rule for field direction

• Motor effect: F = BIL; Fleming's left-hand rule

• Electromagnetic induction: changing flux induces emf

• Transformer: step-up and step-down; turns ratio

Teaching Methodology (Step by Step)

1. Show a compass deflecting near a wire: the Oersted moment.
2. Demonstrate the motor effect with a simple wire and magnet.
3. Move a magnet through a coil connected to a galvanometer.
4. Explain the transformer with the turns-ratio formula.

Classroom Activities

• Apply Fleming's left-hand rule to five given situations.

• Draw a labelled diagram of a transformer.

Assessment Questions

1. State Fleming's left-hand rule.
2. What is electromagnetic induction?
3. A transformer has 100 primary and 500 secondary turns. What type is it?
4. Give one use of a step-down transformer.

Homework

• Write the working of a transformer with diagram.

• Solve numericals on the turns ratio.

Time Allocation

6 periods of 40 minutes each

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Unit 7: Basic Electronics

Learning Objectives

• Distinguish conductors, insulators and semiconductors.

• Explain n-type and p-type semiconductors.

• Describe the p-n junction diode and its uses.

• Explain basic logic gates (AND, OR, NOT).

Key Concepts

• Energy bands idea: conduction made simple

• Intrinsic vs extrinsic semiconductors

• Doping: n-type (electrons) and p-type (holes)

• p-n junction: forward and reverse bias

• Diode uses: rectification; LED idea

Teaching Methodology (Step by Step)

1. Sort materials into conductors and insulators from samples.
2. Explain doping with the 'extra electron' story.
3. Draw the p-n junction in forward and reverse bias.
4. Test a diode's one-way conduction with a simple circuit.

Classroom Activities

• Draw the p-n junction in both biases with labels.

• List five electronic devices and their semiconductor parts.

Assessment Questions

1. What is doping?
2. Distinguish between n-type and p-type semiconductors.
3. Draw the symbol of a diode.
4. What happens in reverse bias?

Homework

• Write the working of a p-n junction diode.

• Draw truth tables of AND, OR, NOT gates.

Time Allocation

5 periods of 40 minutes each

Unit 8: Atomic and Nuclear Physics

Learning Objectives

• Describe the structure of the atom and nucleus.

• Explain radioactivity and its types (alpha, beta, gamma).

• Describe nuclear fission and fusion.

• Discuss peaceful uses and hazards of nuclear energy.

Key Concepts

• Nucleus: protons, neutrons; atomic number, mass number, isotopes

• Radioactivity: alpha, beta, gamma and their properties

• Half-life concept

• Fission (reactors) vs fusion (sun)

• Uses: medicine, power; safety measures

Teaching Methodology (Step by Step)

1. Revise the atom, then zoom into the nucleus.
2. Compare alpha, beta, gamma in a table: charge, penetration.
3. Explain half-life with the halving-squares activity.
4. Debate: nuclear power - benefits vs risks.

Classroom Activities

• Plot a half-life decay curve from given data.

• Research: one medical use of radioisotopes.

Assessment Questions

1. Name the three types of radioactive rays.
2. What is half-life?
3. Distinguish between fission and fusion.
4. Give two peaceful uses of nuclear energy.

Homework

• Write properties of alpha, beta and gamma rays.

• Write a paragraph on uses of radioisotopes.

Time Allocation

5 periods of 40 minutes each

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